

Used Car Price Prediction Using Machine Learning: A Comparative Study of Ridge Regression, Random Forest, and XGBoost

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Abstract

Predicting a fair price for a used car in India is harder than it sounds. Prices change depending on brand, engine specs, age, mileage and location, and there is no single trusted reference for buyers or sellers. This paper compares three models on 6,019 Indian used car listings from Kaggle. Ridge Regression, Random Forest and XGBoost. All three were trained on cleaned and feature-engineered data with a logtransformed price target, then tuned using GridSearchCV. XGBoost produced the best results with $R^2 = 0.9524$, MAE = 1.37 Lakhs and MAPE = 13.69%. SHAP analysis confirmed engine power and vehicle age as the main drivers of prices.

Keywords: Used Car Price Prediction, Machine Learning, Ridge Regression, Random Forest, XGBoost, SHAP, GridSearchCV, Feature Engineering.

I. INTRODUCTION

The Indian used car market has grown considerably over the last decade, but pricing is still inconsistent. Without a reliable reference point, buyers often overpay and sellers undervalue their vehicles. This project builds a machine learning pipeline to predict used car prices using 6,019 listings from Kaggle, comparing Ridge Regression as a linear baseline, Random Forest as an ensemble approach and XGBoost as a gradient-boosted model. The aim is to identify which model is most accurate and which features drive the predictions.

II. RELATED WORK

Research on used car price prediction consistently shows two things: log-transforming the target variable helps and tree-based models outperform linear ones. Gegic et al. (2019) found this on Bosnian data, and Pal et al. (2018) confirmed it on Indian OLX listings where Random Forest reached R^2 above 0.85. Both studies noted that raw price distributions are heavily right-skewed, making standard linear regression unreliable. XGBoost, developed by Chen and Guestrin (2016), comes with L1/L2 regularisation and column subsampling built in and performs strongly on structured regression tasks. Lundberg and Lee (2017) introduced SHAP values, which have become the standard tool for explaining model predictions, particularly in applications where the reasoning behind an estimate needs to be communicated to end users.

III. METHODS

The pipeline shows eight stages from raw data to final evaluation, as shown in Figure 1.

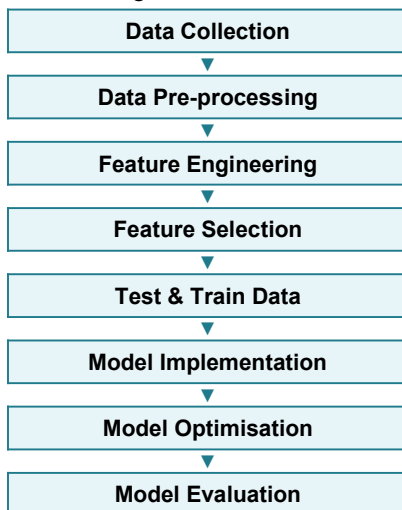


Figure 1: Framework of the proposed methodology.

Data Collection

The dataset was sourced from Kaggle (Kaggle, 2023) and contains 6,019 used car listings from the Indian market. Each listing includes brand name, city, registration year, kilometres driven, fuel type, transmission type, owner type, mileage in kmpl, engine capacity in cc, power in bhp, number of seats and the sale price in Lakhs as the target variable. One Lakh equals 100,000 Indian Rupees. A new price column was also present but was dropped early on because 86% of its values were missing.

Data Cleaning and Pre-processing

Three columns came in as unusable mixed strings, for example '1500 cc' or 'null bhp'. These were converted to numeric values using `pd.to_numeric` with errors set to `coerce` which turns anything unparseable into a NaN. Residual missing values in Engine, Power, Mileage and Seats were all under 1% of the dataset and were filled with column medians, which are more robust than means when distributions are skewed.

One listing was impossible to ignore: a 2017 BMW with 6,500,000 kilometres recorded. That figure would require covering roughly 2,000 km every day for eight and a half years straight. It was clearly a data entry mistake and was removed from the dataset.

Exploratory Data Analysis

The target variable had a skewness of 3.34, which would cause a linear model to weight high-value outliers too heavily. The correlation heatmap in Figure 2 showed that Power_(bhp) and Engine_(cc) have the strongest numerical relationship with price at $r = 0.77$ and 0.66 . These two features are also highly correlated with each other at $r = 0.86$, which creates a multicollinearity problem. Ridge Regression was chosen as the linear baseline partly because its L2 regularisation is well suited to handling correlated predictors.

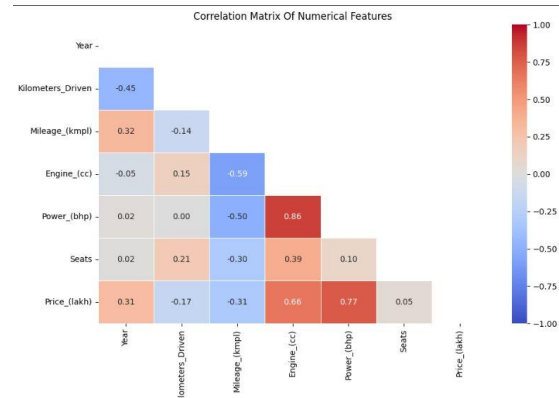


Figure 2: Correlation matrix of numerical features. Power_(bhp) and Engine_(cc) are the strongest predictors of price.

Feature Engineering

Brand was extracted as the first word of the Name column, with Land Rover handled as a two-word exception. Rather than keeping the raw registration year, vehicle age was computed as 2026 minus Year, which captures depreciation more directly. Kilometres driven was log-transformed to reduce its skew. The target price column was also log-transformed, bringing skewness down from 3.34 to 0.75, which makes the regression task more realistic for all three models.

Feature Selection

The final input set contained 11 features: six numerical ones being Mileage, Engine, Power, Seats, Age and Km_log and five categorical ones being Name, Location, Fuel Type, Transmission and Owner_Type. No automated filter method was applied. Tree-based models deal with irrelevant features naturally through how they split data and Ridge's L2 penalty shrinks weak coefficients toward zero rather than cutting them out entirely. Both approaches manage uninformative features without needing a separate selection step.

Splitting Data

The data were split into 80% for training and 20% for testing, giving 4,814 training samples and 1,204 test samples. A fixed random seed of 42 was used throughout to make the results reproducible. The test set was kept completely separate and only used once, for the final performance measurement after all tuning was complete.

Scaling the Data

All six numerical features were standardised using StandardScaler. This is important for Ridge Regression since L2 regularisation penalises coefficient magnitudes, meaning features on larger numeric scales would be disproportionately shrunk without it. Tree-based models do not require scaling, but applying it consistently keeps the comparison fair.

Preprocessing Pipeline

A ColumnTransformer applied StandardScaler to numerical features and OneHotEncoder to categorical ones, all wrapped in a Pipeline object. This ensures all transformations are fitted only on training data within each cross-validation fold, preventing any information from the test set leaking into the model (Pedregosa et al., 2011).

Model Implementation

Ridge Regression: A regularised linear model where the L2 penalty term controls how much large coefficients are shrunk. Higher alpha values produce more shrinkage. Predictions were generated in log-price space and converted back to Lakhs using np.expml before evaluation.

Random Forest: An ensemble of decision trees where each tree is trained on a different bootstrapped sample of the data. Predictions are averaged across all trees, which reduces variance compared to a single tree. The same back-transformation from log-price was applied before computing metrics.

XGBoost: A gradient boosting algorithm that builds trees sequentially, with each tree fitted to the residuals left by the previous ones. L1 and L2 regularisation along with row and column subsampling are used to keep the model from overfitting. It has a strong track record on tabular regression problems (Chen and Guestrin, 2016).

Model Optimisation

GridSearchCV with 5-fold cross-validation was used to tune all three models, scoring on R2. For Ridge, six alpha values were tested producing 30 fits in total. The best alpha was 1.0. For Random Forest, the grid covered n_estimators, max_depth, max_features and min_samples_split across 24 combinations, giving 120 fits. The best setup was n_estimators = 200, max_depth = None and max_features = 0.5. For XGBoost, 24 combinations of n_estimators, max_depth,

learning_rate and subsample were tested, again 120 fits. The best configuration was n_estimators = 200, max_depth = 7, learning_rate = 0.1 and subsample = 0.8.

Model Evaluation

Performance was measured on back-transformed predictions in Lakh units using four metrics: R2 for variance explained, MAE for average prediction error in Lakhs, RMSE which penalises large errors more heavily, and MAPE which gives a percentage error independent of price scale. Cross-validation R2 on the training set was also recorded to assess generalisation stability.

Feature Importance

For Ridge, feature importance was read from the magnitude of the model coefficients on the log-price scale. For Random Forest and XGBoost, Gini impurity-based scores were used. For XGBoost specifically, SHAP values were also computed to give a more complete picture of how individual features affect each prediction.

IV. RESULTS AND DISCUSSION

Baseline Model Performance

All three models performed well on the test set without any tuning. XGBoost achieved R2 = 0.9462 with MAE = 1.49 Lakhs. Random Forest came in at R2 = 0.9407 and MAE = 1.55 Lakhs. Ridge trailed at R2 = 0.9224 and MAE = 1.76 Lakhs, with a MAPE of 19.66% compared to around 15% for the other two. Cross-validation standard deviations were between 0.011 and 0.014 across all three, which shows they each generalise consistently.

Table 1: Baseline Model Performance

Model	R ²	MAE (Lk)	RMSE (Lk)	MAPE (%)	CV R ²
Ridge Regression	0.9224	1.76	4.56	19.66	0.9229±0.0110
Random Forest	0.9407	1.55	4.87	15.84	0.9361±0.0141
XGBoost	0.9462	1.49	4.13	15.54	0.9419±0.0123

Optimised Model Performance

After tuning, XGBoost improved the most: R2 rose to 0.9524, MAE dropped to 1.37 Lakhs and MAPE fell to 13.69%. Random Forest improved slightly to R2 = 0.9444. Ridge stayed exactly the same. The grid search confirmed alpha = 1.0 was already optimal, and a linear model on data this non-linear has a natural ceiling it cannot go beyond.

Table 2: Optimised Model Performance

Model	R ²	MAE (Lk)	RMSE (Lk)	MAPE (%)
Ridge (Optimised)	0.9224	1.76	4.56	19.66
RF (Optimised)	0.9444	1.54	4.80	15.43
XGBoost (Optimised)	0.9524	1.37	4.69	13.69

Actual vs Predicted and Residual Analysis

Figure 3 shows the Ridge actual versus predicted scatter. Points sit close to the diagonal for lower-priced cars, but the scatter widens noticeably above 60 Lakhs. The residual plot in Figure 4 makes this clearer: the spread of residuals increases as predicted price increases, which is a sign of heteroscedasticity. This is a known limitation of linear regression on price data, even after log-transformation. Ridge does a reasonable job on ordinary cars; for premium vehicles it becomes less reliable.

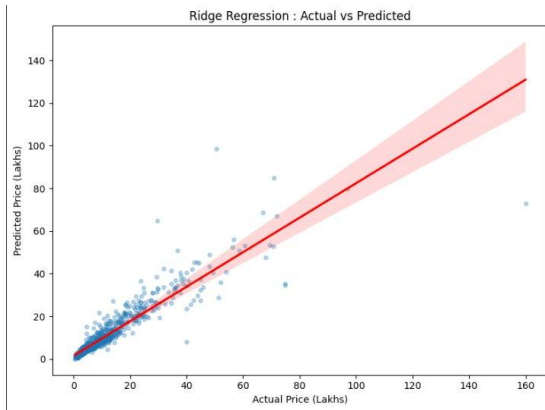


Figure 3: Ridge Regression : Actual vs Predicted (Lakhs). Tight at lower prices, increasing spread at higher values.

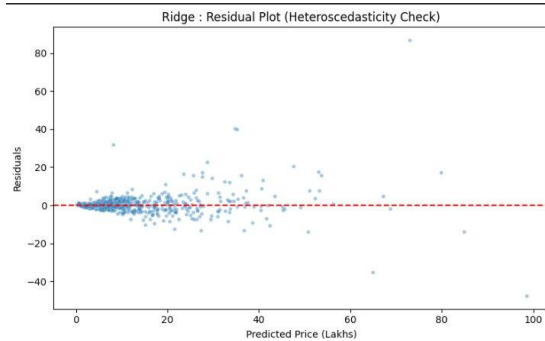


Figure 4: Ridge Regression residual plot. Funnel shape confirms heteroscedasticity, prediction error grows with price.

Feature Importance Analysis

Looking at Ridge's coefficients in Figure 5, brand name is the dominant signal. Mini, Land Rover, Jaguar, Mercedes-Benz, BMW and Audi all carry large positive coefficients, while Tata, Datsun and Chevrolet pull predicted prices down. Electric fuel type also appears near the top. For a linear model this makes sense: brand is a clean categorical variable that translates well into a single coefficient.

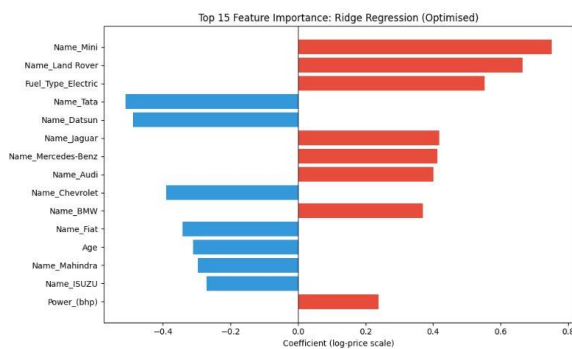


Figure 5: Top 15 Ridge Regression coefficients. Premium brand names and Electric fuel type are the dominant positive predictors.

Random Forest and XGBoost give a noticeably different picture. Both put Power_(bhp) at the top by a clear margin, as shown in Figures 6 and 7, with Engine_(cc) and Age following behind. Tree-based models can capture the direct, continuous relationship between engine power and price. Ridge cannot do this as cleanly because it has to approximate that relationship through brand and fuel-type dummies, which is probably why the two ensemble models do better at the premium end of the market.

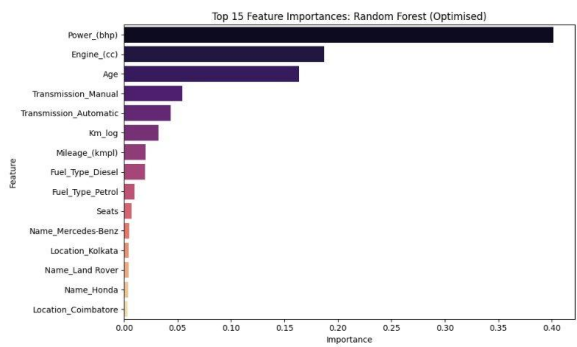


Figure 6: Random Forest top 15 feature importances. Power_(bhp) dominates, followed by Engine_(cc) and Age.

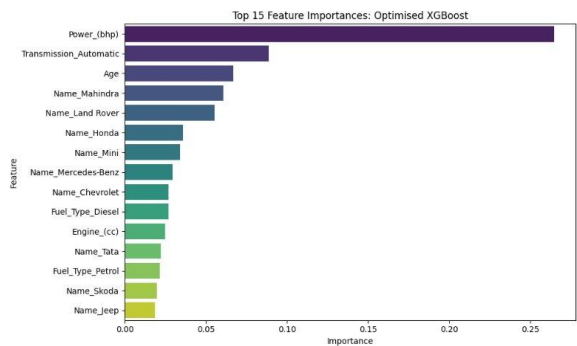


Figure 7: XGBoost top 15 feature importances. Power_(bhp) leads; brand names (Mahindra, Land Rover) also contribute.

V. MODEL EXPLAINABILITY: SHAP ANALYSIS

SHAP values break each prediction down into contributions from individual features, based on Shapley values from game theory (Lundberg and Lee, 2017). What makes SHAP more useful than a standard importance score is that it shows direction: you can see not just that Power_(bhp) matters but whether a particular value of it is pushing the prediction up or down.

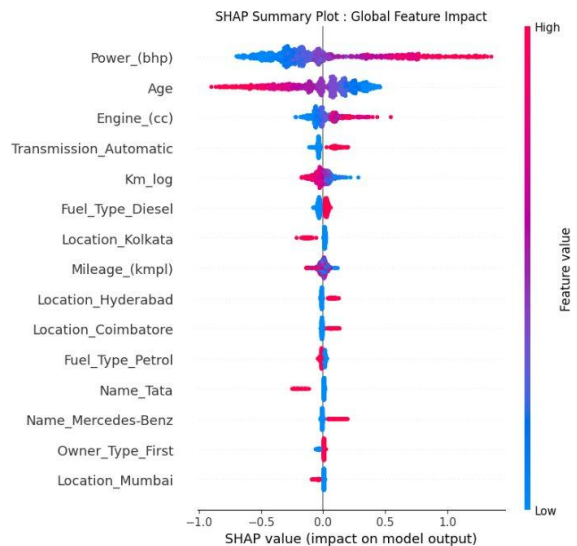


Figure 8: SHAP beeswarm plot for optimised XGBoost. Red = high feature value, blue = low. Width of spread reflects global importance.

Figure 8 shows Power_(bhp) with the widest spread of any feature. High-power cars sit well to the right, meaning they are predicted at higher prices and low-power cars sit to the left. Age behaves in the opposite direction, older vehicles consistently pull the predicted price down, which matches how depreciation works in practice. Engine_(cc) has a moderate positive effect, and Km_log reduces predictions for high-mileage cars as expected. Location variables for Kolkata, Hyderabad and Coimbatore show small consistent

downward adjustments, which reflects regional price differences across India. These patterns are all intuitive, which is a good sign that the model has learned genuine market behaviour rather than noise in the training data.

VI. ETHICAL, LEGAL, AND PROFESSIONAL CONSIDERATIONS

Data Ethics and Privacy

The dataset is publicly available on Kaggle and contains no personal information about individuals (Kaggle, 2023). For this research context, no ethical concerns arise from the data itself. Any organisation looking to deploy this model commercially would need to ensure compliance with the UK GDPR and Data Protection Act 2018, particularly where automated pricing decisions could materially affect individuals under Article 22.

Algorithmic Fairness

The SHAP results show that location has a measurable effect on predicted prices. Whether this reflects genuine differences in regional market conditions or biases embedded in historical listing data is not something that can be determined from the dataset alone. Before deploying this model to set prices in practice, it would be worth auditing it for geographic disparate impact to ensure sellers and buyers in certain cities are not being systematically disadvantaged.

Model Transparency and Explainability

Using a complex model to value something as significant as a vehicle purchase, without any way to explain why a particular estimate was produced, is increasingly hard to justify. The EU AI Act (2024) identifies price prediction systems as potentially high-risk and requires that explainability mechanisms are in place. Including SHAP in this pipeline is not just an academic addition; it is the kind of explanation layer that would be required before a model like this could be used responsibly in a commercial setting.

Professional Standards

This project was carried out in line with the BCS Code of Conduct. All data sources are properly cited, limitations are reported openly, and no claims are made beyond what the results actually support. The heteroscedasticity identified in Ridge's residuals, the fact that the dataset is specific to the Indian market, and the absence of information like service history and accident records are all acknowledged as constraints on how broadly these findings apply.

VII. FUTURE WORKS AND CONCLUSION

XGBoost was the strongest model across all metrics after tuning, reaching $R^2 = 0.9524$, $MAE = 1.37$ Lakhs and $MAPE = 13.69\%$. Random Forest performed well but fell slightly short. Ridge Regression explained 92% of price variance, which is not bad for a linear model, but it struggled at the premium end of the market where the relationship between features and price becomes more non-linear.

Possibly the most practically useful part of the project is the SHAP analysis. It shows that the model's predictions are driven by factors that align with how buyers actually think about car value: engine power, how old the vehicle is, and engine size. To test end-to-end functionality, a 5-year-old Maruti Petrol Manual with 45,000 km and 82 bhp was run through the pipeline and came out at 5.45 Lakhs, which is a reasonable estimate for that kind of car in the Indian market.

There is room to develop this further. Adding service history, accident records and more recent listings would improve accuracy and relevance. Temporal modelling could account for market price drift over time, and the regional pricing differences flagged by SHAP would be worth investigating if the model were ever deployed across different cities.

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